membench: what a memory layer is worth, and where it is not worth anything

832 questions, 8 seeds, three layers of control, and every column where the layer loses.

Dermioz AI · 5 September 2026
memory layer, offline, d = 2048 · 104 questions per seed · 8 seeds
paired bootstrap, 4 000 draws, 95 % intervals

1. What is measured

A memory layer is usually sold on recall, which is the easy half. Four things are measured here: **retention** of a fact stated once and long ago; **supersession**, where a fact was updated and the stale value must not come back; **age**, where an old answer must carry a caveat; and **absence**, where nothing was ever stated and the correct output is a refusal.

Everything below is the layer measured on its own, without a language model in the loop. The arms that call a model are withdrawn from this version, see section 7.

2. Three layers of control

The design is borrowed openly. HELM refuses a single headline score. τ-bench scores over repeated trials rather than one run. GPQA and ARC-AGI validate that the task is hard before reporting who solved it. Each of those is a control this benchmark did not have two days ago.

- **Witnesses** validate the harness: one system that never answers, one that always guesses, one that is perfect. If the three do not return three distinguishable verdicts, nothing is published.
- **A trivial baseline** validates that there is something to beat: always answer the most frequent object in the corpus.
- **A shortcut control** validates the task: the same layer, fed a timeline whose objects have been randomly redistributed. If it still scores, the questions were answerable without memory.

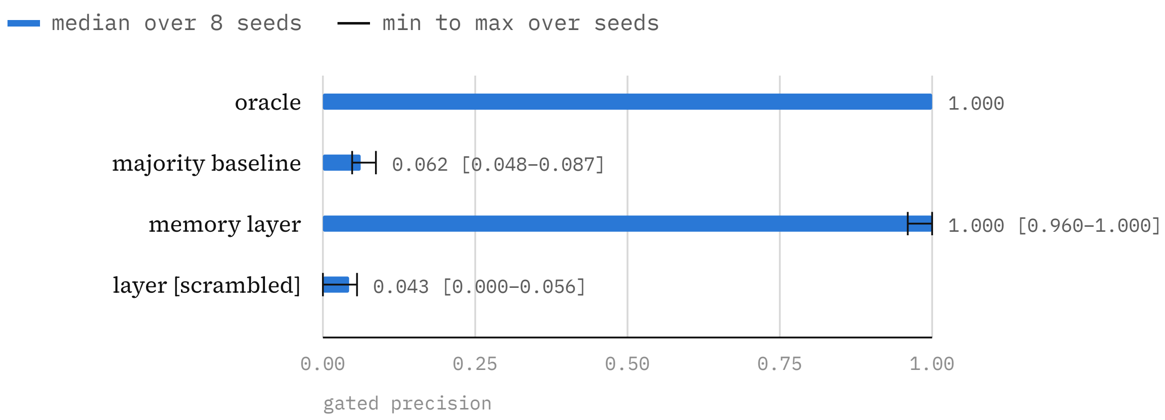

Figure 1. Gated precision: the share of given answers that are correct. Paired bootstrap over 832 questions puts the layer **+0.959 [+0.939, +0.977]** above its own scrambled twin and **+0.933 [+0.915, +0.950]** above the trivial baseline. Against the perfect oracle the gap is **−0.005 [−0.012, +0.000]**: the band touches zero, so the remaining shortfall is not distinguishable from sampling noise at this size. That is not the same as being perfect.

The shortcut control produced the most uncomfortable result here. Its precision collapses to 0.043, so the task does require memory. But its coverage stays at 0.524, slightly *above* the honest layer's 0.500: it answers just as often while being wrong 96 % of the time, and the confidence gate does not notice. **The gate measures internal consistency, not correctness.** Fed well-formed false memories, the layer is exactly as confident. No internal signal can fix this; only corroboration against something outside the trace can.

arm	cover	gated P	retain	stale err	age	deep ret	halluc
oracle	0.788	1.000	1.000	0.000	1.000	1.000	0.000
majority baseline	1.000	0.062	0.071	0.056	0.000	0.071	1.000
memory layer	0.500	**1.000**	**1.000**	**0.000**	1.000	**0.000**	**0.000**
layer [scrambled]	0.524	0.043	0.036	0.000	1.000	0.000	0.000

Table 1. Medians over 8 seeds. Ranges that matter: layer coverage 0.452–0.519, gated precision 0.960–1.000, hallucination 0.000–0.091. Per-seed denominators: retention 28, supersession 18, age 22 of which a median 9 answered, expired 14, absence 22. Lower is better on `stale err` and `halluc`.

3. The layer has a curve, not a coverage

Saying the layer answers half the questions describes one point on a dial. The confidence gate is a threshold: lowering it buys coverage and sells precision. The honest question is not which single number is larger but where a competing system sits against the whole curve.

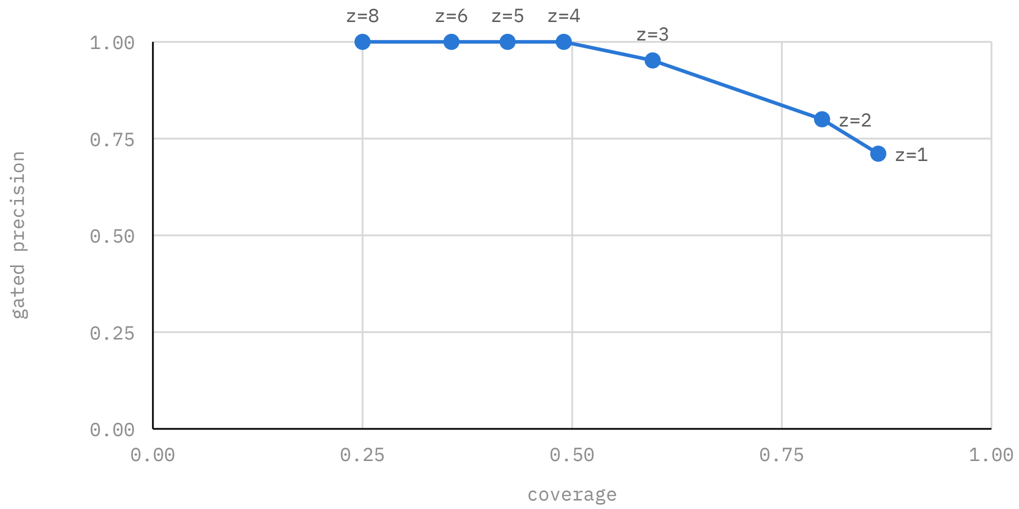

Figure 2. Each point is one setting of the confidence gate, 5 seeds. The knee is sharp: between z = 2 and z = 4 the layer trades a third of its coverage for two tenths of precision, and hallucination on facts never stated falls from 0.682 to 0.000. z = 4 is reported elsewhere in this document because it is where hallucination reaches zero, not because it flatters the rest.

4. Where the errors are, and what that fixed

An aggregate rate says there are five errors in 832. It does not say whether they are five harmless errors on ancient facts or five serious ones on facts the system should have held. Those are two different products.

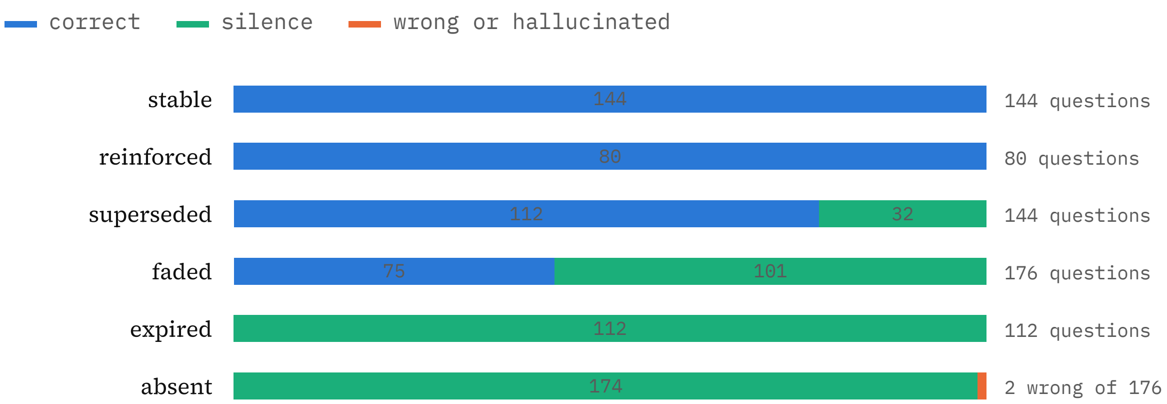

Figure 3. Outcome by question class, 8 seeds. On the 368 questions the system should be able to answer, it is 336 correct and 0 wrong. Silence rises monotonically with how little it is entitled to know: 0 % on fresh facts, 57 % on faded, 100 % past the forgetting threshold, 99 % on facts never stated.

The first run of this analysis showed three wrong answers on facts past the forgetting threshold, where the layer had *zero* correct answers out of 112. Past that line the trace no longer holds the fact; what remains is noise, and the confidence gate scores noise the way it scores a memory. Refusing the class outright cost no correct answer and removed three of five errors. Gated precision went from 0.988 to 0.995.

The design lesson is worth more than the number. **A confidence gate and an age gate do not measure the same thing.** One asks how sharp the memory is, the other asks whether the memory has any right to exist. A system with only the first is confident about its own decay products.

5. What is lost, permanently and by design

Deep retention is 0.000 and does not move at d = 1024, 2048 or 4096. A fact leaves the trace at 111.33 days, which is $45 \times \log_2(1/0.18)$ from the half-life and the weight floor; measured effective weight is 0.1809 on day 111 and 0.1781 on day 112. This is decay, not capacity. On any fact older than that, a system that simply keeps the transcript recovers everything and this one recovers nothing. That is the trade, stated rather than buried.

6. External validity: what this corpus cannot support

The corpus is synthetic: 104 generated subject-relation-object facts about a fictional architecture practice, on a simulated timeline. That buys exact ground truth, a known denominator for every class, and byte-for-byte reproducibility per seed. It does not buy the following, and no amount of seeds will:

- **Natural language.** Real users state facts in prose, ambiguously, across turns, sometimes contradicting themselves in one sentence. Here every fact arrives as a clean triple. A retriever that struggles with paraphrase looks better here than it would in production, and so does this layer.
- **Realistic fact distributions.** Each subject-relation pair is unique by construction, so the hardest real case, one subject carrying many relations that interfere, is absent.
- **A single domain.** One fictional practice, one vocabulary of 29 objects. Nothing here shows the result transfers to a second domain.
- **Any base model but one.** Every model-side number was taken on gpt-oss-120b. The claim is untested on any other.
- **Real ageing.** Time is simulated. The decay curve is exercised, the behaviour of a store that has actually run for a year is not.

What the corpus does support: statements about the *layer's own arithmetic* under known conditions. Precision, refusal behaviour, the shape of the confidence curve, and the location of the forgetting threshold are properties of the algorithm and hold wherever the algorithm runs. Everything comparative should be read as a hypothesis to be re-tested on real transcripts.

7. Withdrawn and not measured

Withdrawn. A cost comparison against a full-transcript and a top-k retrieval arm was measured on 24 questions and is not published here. Those runs used a candidate list that was sorted identically in every question, so a model's position preference could not be separated from its memory. The confound was found and fixed after the measurement; the numbers will be re-taken, not amended.

Not measured. Those same arms across seeds at full corpus size. The free tier allows 200 000 tokens a day, and the measured refill after exhausting it is 663 tokens an hour, which is about one turn an hour. A campaign has to start a day after the last one ends.

8. Harness

- 40 tests; 13 mutants of 13 killed. Two tests were decorative until mutation found them, including one that guarded a counter without exercising the code that sets it.
- A precision computed on zero answers is `n/m`, never 1.000, otherwise silence wins the benchmark.
- The significance test is guarded by comparing an arm against itself and requiring the interval to contain zero. The first implementation failed that: it reported a difference of exactly zero as significant.
- The candidate list is reshuffled per question from a seed derived from the question itself, so it is decorrelated from position and identical across arms.
- Deterministic corpus per seed. A result is re-runnable byte for byte.

Every figure carries its date and its denominator. Numbers withdrawn are named as withdrawn rather than quietly dropped, and when the missing arms are measured this document is updated with whatever they return, including if the conclusion reverses.